"Physics as Information Processing" ~ Chris Fields ~ Lecture 4

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Hello and welcome everyone. This is Physics as Information Processing, Session 4 on Communicating Observers.

It's the 10th of August, 2023. Chris, looking forward to this lecture. Off to you.

Thank you, Daniel, and welcome to this session. As Daniel said, this is about communicating observers and fasten your seatbelts because we have a lot to cover in the next hour.

So just to do a quick review, we've been talking about quantum information theory, and we've been talking about it in the simplest possible setting of two systems, Alice and Bob, that exchange information through their mutual boundary.

And we've assumed that everything is finite dimensional, so we don't need to worry about infinite energy transfer across this boundary.

And, of course, we assume conservation of energy, so there are no sources or sinks in either of the systems, Alice and Bob.

And we emphasize that this way of depicting things is topological. We're not assuming any background geometry, no sort of spatial embedding, which is going to become very important in this session.

In the last session, we talked about how the boundary, which I'll always call B, can be thought of as a channel that comprises n qubits that are independent.

And Alice and Bob each can prepare and measure those qubits.

And they can always choose independently their reference frames for doing so, so how they define up and down when they're measuring a spin with effectively a Z-spin operator to interact with these qubits.

We talked a little bit about the free energy principle and its quantum formulation.

And the principle basically says that interacting systems will behave in a way that aligns their reference frames and that they therefore asymptotically approach entanglement.

So the FEP is a classical limit of the principle of unitarity.

And that's one reason for studying quantum information to learn more about the FEP and so to learn more about active inference and what the theory of active inference says about agents.

And last time we talked a lot about the constraints that are imposed by a limited free energy supply on any agent.

Recall that we talked about the fact that in the absence of sources of free energy within an agent.

So as long as you conserve energy globally, that the free energy that each agent uses has to come from its environment.

That's clearly true for us.

We have to consume oxygen and eat food and things like that to keep running as systems.

And we discussed a little bit this theorem that if two reference frames or quantum processes, Q1 and Q2, don't commute,

they have to be implemented by compartments that communicate classically.

And we'll use this extensively today.

But the bottom line last time was that physics alone gives us compartmentalization in systems whenever the systems are complex enough to have multiple kinds of measurements that they can't deploy simultaneously.

So where we're going today, we're going to start from this place we left off last time of the possibility of agents that are compartmentalized and into compartments that have to communicate classically.

So we're going to talk about agents that can be compartmentalized into multiple subagents that share an environment, the environment consisting of Bob, and how those agents talk about that shared environment.

So the motivation for this is simple.

It's that what we do all the time is a group activity.

And what we're doing right now is a group activity that involves sharing an environment, what's shown on this screen, and talking about it.

And the whole post-session discussion process will be another example of just that sort of thing.

So to talk about science at all, or to talk about any kind of social relations, or to talk about anything other than solipsism, we have to be able to talk about how multiple agents communicate about some shared environment.

And so that's what we're going to develop a formal way of talking about.

And there are two things to notice right off the top about this.

One is that the way we've defined interactions at boundaries, they're effectively instantaneous.

So each agent, remember, interacts with its boundary with some set of operators.

And what those operators do is effectively measure, each operator effectively measures the spin of a qubit.

So this is a very fast process.

The other thing that we then need to notice is that if we have two agents and they're communicating with each other classically,

then they're going to be using degrees of freedom of their shared environment in order to communicate.

And that's exactly what we're doing right now.

For example, we're using degrees of freedom of the internet to communicate.

I'm using degrees of freedom of the local atmosphere to talk to my microphone.

And I'm using degrees of freedom of the local photon field to see my screen.

So all communication between agents, and certainly all classical communication between agents, uses degrees of freedom of the environment.

And the environment, remember, is the other system, Bob, that's on the other side of that shared boundary.

So we're going to be interested in how components of Alice use Bob as a communication channel, and in fact, as multiple communication channels.

And I just wanted to quote my colleague and friend, Mike Levin, in his statement that all intelligence is composite.

All of us, all organisms, even single cells, are composite systems that comprise many, many different agents that are able to make different kinds of measurements

and execute different kinds of actions and so have to communicate classically.

So we're talking about all of life here, as well as all of social systems and ecosystems and such things as that.

So the first thing I want to do today is simplify the notation so that we don't have to draw quite such complicated diagrams.

So the picture on the left here is a picture from the last session.

It shows the structure of a very simple agent that makes some measurements using a reference frame called E,

and then writes some results of those measurements to memory using a reference frame called Y.

And those two processes are connected by an internal clock that ticks between the input process and the output process.

And the clock tick corresponds to the expenditure of free energy to write information irreversibly on the boundary.

So the memories are actually written on the boundary because that's where classical information lives in this theory.

So you can think of this whole process from E through the clock and back to Y as a single quantum process or as a quantum computation that maps some sector E of the boundary to some sector Y of the boundary.

And so we can represent it in this much simpler way that's over on the right side as some process Q that maps some sector on the boundary to some other sector on the boundary.

And formally, that process can be thought of as a topological quantum field theory.

Topological here is important.

It's not geometric.

It doesn't assume an embedding in space.

And this will become critically important later on.

And the operation of that TQFT requires a clock tick or it incorporates a tick of this internal clock.

And we can do this because we were able to prove, Jim Glazeberg and Tonino Marciano and I were able to prove last year, that we can always represent a quantum reference frame as a topological quantum field theory.

And this just illustrates the two kinds of transformations between quantum reference frames that are important.

We can take one and divide it into two pieces that commute.

Or we can take a two-part reference frame and we can swap in another part that doesn't commute with the part that we've swapped it in for, but that does commute with the remainder of the reference frame.

So here, R is the reference component of some object identification system.

And we can swap in some different way of looking at the object once we've identified it.

So, for example, if the object that we're identifying is a box that can measure both Z-spin and X-spin, then P and Q can be the Z-spin and X-spin settings on that box that allow us to effectively rotate the filter from this way to that way,

so that we measure vertical spin versus horizontal spin.

And that's really it.

I mean, those are the ways that one can transform quantum reference frames.

Of course, we can do arbitrarily many compositions of this kind of transformation if you do something very complicated.

But they map very simply to TQFTs with the structure.

And you can construct categories out of this and find a functor that's well-defined that maps one category to another.

So it's an example of a nice use of categorical techniques for solving a problem that looks difficult, but in fact, it's very straightforward.

So using this simplified notation, it becomes very easy to see that if we have two systems, Alice and Bob, on opposite sides of this boundary,

and we perfectly align their quantum reference frames, which is what the FEP says they will tend to do,

then they become entangled.

And you can see that they're entangled by seeing that once the reference frames are perfectly aligned,

they jointly implement one quantum process.

And jointly implementing one quantum process just means you're entangled.

The two sides no longer have conditionally independent states.

If one quantum process, which is time-reversible, information-preserving, is implemented jointly by the two systems.

So a lot of things like this become very simple when we think of these processes as just quantum computations going on inside some system,

implemented by the two systems, and how those processes relate across some boundary.

So that's what we're going to do for studying communication today.

Another thing to recall from a couple of sessions ago, or last session, is that if two reference frames, Q1 and Q2, commute,

we can always join them together to form a single reference frame.

And this is just a graphical depiction in this simpler language of how that works.

And topologically, there's no distinction between the left side and the right side.

What I've done is just squeeze the top two green circles together and the bottom two green circles together into one circle.

And that just makes a simple process connecting the top one to the bottom one.

So what we're going to be interested in is situations where this doesn't happen,

where Alice consists of two components that execute reference frames that don't commute.

And so the two components are conditionally independent.

They do each have free choice of reference frame.

And so they have to communicate classically.

They don't share a single quantum process.

So they're not entangled.

And before we go ahead to do that, I just wanted to remind you of some of the no-go theorems that follow from this situation

depicted on the right, where we have a single agent executing a single reference frame,

a single process, a single kind of measurement,

any set of commuting measurements that can be combined into one measurement.

And here are some things Alice can't do.

And we've talked about some of these previously.

So this is basically a reminder.
But Alice can't, using just one measurement,
measure the entanglement entropy across her own boundary.
So Alice can't know that she has an independent state.
She can't know that she's not entangled with Bob.
And that's a useful thing to think about philosophically,
as it applies to you and me.
We can't actually know that we have independent states.
Alice can't measure her own entanglement entropy.
So she can't know whether she has components that have to communicate classically.
She can't measure Bob's entanglement entropy.
So she can't know whether Bob has compartments that have to communicate classically.
And if you think in terms of perception theory,
this says Alice can't know whether Bob consists of separate objects that behave classically.
their behavior can look classical to her.
But she's not able to determine that their behavior is classical,
because she can't measure Bob's entanglement entropy.
She can't measure the dimension of her own boundary.
And recall from a couple of times ago that the boundary includes some sector that supplies free energy
that Alice can't extract information from for further processing.
She can only extract free energy from.
So she can't measure the dimension of that sector.
And so she can't measure the dimension of the whole boundary itself.
And this, of course, means that she can't measure her own dimension either.
And, of course, she can't measure Bob's dimension,
because the only information she can get about Bob
is the classical information that's encoded on this boundary.
She can't reach past the boundary to see how large a system Bob is.
And that's clearly connected to this first point,
that she can't measure the entanglement entropy across Bob,
because the entanglement entropy can only be small
if her dimension and Bob's dimension are both much larger than the dimension of the boundary.
So that's a condition she can't check.
So all of these kind of no-go theorems lead off in interesting directions.
And the only one that we're really going to be interested in here
is the third one, the one about measuring Bob's entanglement entropy,
because we want to talk specifically about how multiple agents,
so multiple components of ALICE,

can access information about Bob's entanglement entropy.
Okay, so recall from the end of the last session
that the last thing we talked about was classical communication.
And we talked about the fact that classical communication
involves assumptions and pitfalls,
and that we have to be very careful whenever we talk about it.
And what we're going to do today is standard practice in physics,
we're simply going to stipulate that these components,
A1 and A2, the separable components of ALICE, communicate classically.
And so we're going to draw it like this.
We're going to draw a classical channel between them.
And that classical channel passes through Bob.
It uses degrees of freedom of the ambient photon field
or the ambient atmosphere or pieces of paper or email
or something like that that's out there in the environment
to exchange classical messages.
And there are a couple of things to note here.
I haven't even written down here.
We're assuming that A1 and A2 share some language
for this classical communication.
That's a big assumption, as we talked about last time.
But I will note explicitly that classical communication takes time.
That's what defines classicality here.
It's causal, so it can't work faster than the speed of light.
This is a topological theory.
We don't have an embedding geometry,
so we can't at this point talk about light,
which is a relationship between space and time,
because we've got time, we've got clocks,
but we don't have any space.
But we can say that whatever passes through this classical channel
requires at least one clock tick on Bob's side.
So it's a process that Bob has to execute,
and executing that process takes time.
So getting the signal from me to you on the internet takes some time,
so the environment's clock has to tick during that.
And all of these assumptions bundled together
turn out to be a fairly serious set of assumptions,

and they involve assumptions about fine-tuning,
about the universe being just right.
And they should make us nervous.
And I just want to leave it at that.
We always assume with complete aplomb
that agents can communicate classically,
but it's not a straightforward assumption.
And it's worth,
any of you who are looking at,
for interesting research projects,
can look into the nature of classical communication.
It's full of hidden assumptions and surprises
and things that aren't at all straightforward,
even though they look like they ought to be.
So there's a challenge.
So why do we assume this?
Why do we assume classical communication?
We do it because we have to,
if we're going to talk about detecting entanglement.
And most of you probably know the history
of the notion of entanglement.
It was first noticed by Einstein, Podolsky, and Rosen,
ever after called EPR in 1935.
And they wrote a paper, which is infamous,
and called the EPR paper,
in which they argued that the phenomenon of entanglement
made no sense and couldn't possibly be physical.
And since quantum theory included this notion of entanglement,
it couldn't possibly be a complete theory
of what's going on in the world.
And Schrodinger replied the same year
with a very detailed paper
where it introduces, among other things,
Schrodinger's cat as a paradox.
And he explains very clearly what entanglement is
and why entanglement follows from quantum theory.
And he uses that discussion to argue
that the classical notion of a physical state

is not actually applicable to the world as it is.
So he answers the EPR paper very directly.
And this discussion
is currently mostly known as the Bohr-Einstein debate
because, of course, Bohr, already in 1928,
had said in print that the classical notion
of causality in space and time
simply doesn't apply anymore.
And Einstein objected to that vigorously.
But that was 1935.
And it was almost 50 years later
that entanglement was first observed in the laboratory.
Here, the 1982 picture is a picture
of the interior of Alan Asbeth's laboratory
when he performed the first entanglement experiment
for which he finally won the Nobel Prize last year.
And these days, this kind of experiment
is done all over the world
and is even done with satellite-borne sources.
This is the key to quantum security,
quantum secured communication,
as we'll discuss in quantum cryptography
and quantum banking, the quantum internet,
all this other stuff that you may have heard of.
So before talking more about these experiments,
I just want to provide a definition of entanglement.
Entangled simply means not separable.
That's the definition.
So a state, a joint state, A, B, is entangled
if and only if it does not factor
into two separable states.
And that's it.
That's the definition.
And the literature, and especially the popular literature,
is absolutely rife with hype
and various kinds of mysterification of entanglement.
And authors trying to convince you
that entanglement is this amazing kind of woo-woo thing.

And whenever you run across that,
go back to the definition.
I mean, entanglement simply means
this simple mathematical condition
of not separability
or of a state that isn't factorizable.
And that means that the state
does not display classical conditional independence.
So classically, that's the way to think about it.
It's a failure of conditional independence.
And it's ubiquitous in quantum theory,
even though it wasn't actually observed
for almost 50 years after it was predicted.
So here's what all these experiments look like.
They're called Bell-EPR experiments,
after John Bell,
who derived the statistical criterion
for detecting entanglement.
And EPR, Einstein, Podolsky, and Rosen,
who did this as a thought experiment
and said,
if this occurs in the theory,
it can't possibly be real.
And of course, it is real,
as now amply demonstrated by experimentation.
Here's how they work.
There's a source.
The source produces a state
that is entangled.
And I've written the typical,
what's called a Bell state.
It's up plus or minus down
divided by the square root of 2
just to normalize it.
But up plus or minus down
is clearly not the same as up times down.
So this is not a separable state.
And this state propagates

outward in space and time
to two observers
who have independent laboratories,
for example, on different continents.
In that space-based experiment
that I showed you the picture of
from the cover of Science
a few years ago, 2017, I think.
The truth observers have freedom
to choose their own reference frame,
their own z-axis for measuring spin.
They do a spin measurement.
And they do this over and over and over again.
They accumulate statistics.
And they exchange their results
so that they can analyze
the joint statistics of their experiment.
And they check to see
whether that joint statistics
is consistent with classical probability theory.
And if it isn't
to a statistically significant level,
then they've been able
to detect entanglement.
Now, notice that this process
requires classical communication.
They have to make
their independent measurements
in their own laboratories,
and then they have to exchange results.
Or one of them has to send results
to the other one,
or they both have to send
their results to some third party.
But in all of these cases,
they're exchanging some classical data.
And so we have the assumption
that whoever gets that classical data

knows the language it was recorded in,
and so can do a joint analysis.
So I'll just show you another picture
of this experiment
that's in a standard space-time diagram.
Here, time is vertical.
Space is horizontal.
The two observers are labeled Alice and Bob.
They share a quantum channel
that goes from Alice to the source to Bob.
That's the entangled state.
They do some processing,
and then one of them here, Bob,
sends the results to Alice.
And that's a classical communication.
It takes time,
and it traverses space.
So this step is classical.
So this is why we have to assume
classical communication
to talk about detecting
or using entanglement.
So now I want to represent this
in a different way
using our little picture earlier.
We're going to think of Alice again
as two components,
A1 and A2.
They're separable,
so they have to communicate classically.
They have a classical channel
that goes through Bob,
and they also now share
some quantum channel.
They share an entangled state.
And the entangled state, of course,
is produced by Bob.
The source is part of Bob.

And their detectors,
if you want to think about it that way,
are located on this boundary B,
where their interface with the source
is on this boundary B.
And each of them executes a process
that accumulates data
from the quantum channel
and communicates it
to the other observer
via the classical channel.
Now this picture illustrates
what's called a LOC protocol.
LOC means local operations
and classical communication.
So A1 and A2
are each operating locally
on their boundary.
They're making measurements
locally on their boundary.
And they're communicating
via actions locally
on their boundary
and observations locally
on their boundary.
So for example,
I'm communicating
with my computer here.
We can consider it my boundary.
And stuff is happening
in the environment
that's implemented
by the internet.
And you're receiving
those communications
locally on your boundary,
on your laptop
or whatever it is you're using.

So that's what
local operations means.
And these protocols
specifically involve
classical communication.
And so they're able
to detect entanglement.
And a Bell EPR experiment
is just an example
of a LOC protocol.
how the observers
exchange information
about how they're going
to do their experiments.
They're both going
to measure Z-spin.
They're going to use
independently chosen
reference frames.
And they're going to look
at the output
from some agreed upon source.
So that's all
classically set up.
Then they actually
make their observations.
and then they
classically communicate
their results.
So classical communication
front and back.
LOC protocols
are not limited
to this sort of
Bell EPR
sort of setting.
We can also represent
a scattering experiment

as a LOC protocol.
So here's a representation
of scattering
as a LOC protocol.
And you can see
all I've done
is cut
that previous
boundary
at the division
between
A1 and A2.
And I've then
extended it
in some
external
time,
the time
that is
counted by
Bob's clock,
not by Alice's
clock.
That's what's
meant by
external.
And I've
relabeled
Alice 2
as Alice 1
at a later
time.
So now
what Alice
is doing
is communicating
with her
past self.

Or A1
is communicating
with her
future self.
And they're
each doing
something different.
A1 is
executing
a procedure
that in
the laboratory
we call
state preparation.
She's preparing
a state
that we'll
call
N,
which is
just the
initial state
of some
scattering
experiment.
So for
example,
at the
LHC,
N
is some
state
of two
protons
that are
quarterling
around the
ring at

the LHC
at close
to the
speed of
light.
And
Alice
then later
on at
T plus
delta T
is doing
a different
kind of
quantum
process.
She's
analyzing
the data
coming out
of something
like the
ATLAS
detector,
which is
a multi-particle
counter
that's located
at the
LHC.
So they
share a
quantum
channel,
which is
the scattering
channel that
takes the

incoming state
and produces
the outgoing
state.
And I
recall from
the very
first session
as Feynman
told us,
to understand
this transition
from in
to out,
we have
to integrate
over all
possible paths
that are
consistent
with
conservation
of
spin
helicity,
basically
momentum
plus spin.
all the
different
spins that
are relevant
in a
high-energy
physics
experiment,
including
things like

lepton
number,
which is
basically a
spin.
Now,
how do we
know?
How do we
know what
we have to
conserve?
We know
what we have
to conserve
because Alice
is doing the
experiment,
has a
classical
memory that
allows her
to remember
what she
prepared.
She can
remember
that what
she did
back in
the past
was set
two protons
counter-rotating
in the LHC.
And they
had certain
energies,

and so they
had certain
momenta,
and they
were protons,
so they
had certain
spin variables.
All of
that has
to be
remembered
to make
any sense
of the
output
of the
detectors.
And so
to actually
come up
with a
measured
state
that
makes
sense.
And very
often this
memory
channel is
neglected
by physicists.
And I'm
going to pause
here and let
Ander ask
a question.

Thank you,
Chris.
So does this
imply that
every time you
have a log
channel,
the quantum
part is
a scattering
picture,
like a
unitary
evolution,
even in
the earlier
EPR
picture?
Is that
quantum
channel
portion
always,
a unitary
evolution
scattering
picture?
Thank you.
Yes,
for
essentially
the reason
that
to find
a
classical
channel,
you're

always
assuming
you have
something
that's
protected
from
decoherence
by the
environment.
So
if you're
doing an
EPR
experiment,
you have
to assume
that you
don't have
an environmental
decoherence
between the
source and
the detectors.
And
similarly,
when you're
doing a
scattering
experiment,
you have to
make the
assumption
that the
environment
is not
reaching in
there and

messing
with the
outgoing state
that you're
going to
see or
reaching in
and messing
with the
incoming state
and changing
it in some
way that
re-prepares
it in a
way that
you don't
know,
that you
have no
knowledge
of.
So yes,
you're making
a fairly
strong
anti-decoherence
assumption
in any of
these settings
when you're
talking about
having a
quantum
channel.
Does that
answer the
question?

Yes,
thank you.
Okay,
great.
So
we can
think of
Locke,
we can
think of
scattering
as Locke,
and I
just wanted
to now
emphasize
that
effectively
what
Locke
is
implementing
is two
parallel
memories,
a classical
memory,
which,
as we
said in
the case
of scattering,
is memory
for preparation.
In the
EPR case,
it's memory
for

preparation
for what
the experimental
setup is,
for what
the instructions
for doing
the experiment
are,
and we
also have
a quantum
channel,
which is
effectively
another memory.
It's something
that's extended
through time,
and again,
it's implemented
by the environment,
implemented by
Bob,
and it
involves
Bob's
clock ticking.
So,
whenever we
have two
memories
in parallel,
we can
construct an
error-correcting
code.
So,

think of a
simple
classical
error-correcting
code,
for example,
the use of
parity bits
in exchanging
digital data.
So,
if I send
you a bit
string,
classically,
and I send
it to you
two or three
times,
so there's
some redundancy
in my
signal,
and I send
you a
parity bit,
I say,
okay,
add up your
bit string,
and the
result should
be one.
It's one
on my
side.
If you
add it up

and you get
zero,
there's an
error someplace.
You can use
that parity
bit to
detect errors.
So,
you can
throw out
the bit
strings that
you receive
that have
parity
zero,
and say,
oh,
the
environment
interfered
somehow with
the signal,
or some
eavesdropper
interfered with
the signal
and made
a mistake,
or something
like that.
So,
this corresponds
to environmental
decoherence not
being a good
assumption,

even though it's
classical.
Environmental
interference is
not a good
assumption in
this case.
So,
a quantum
error correcting
code is
basically the
same idea.
I use
some
feature
of the
fact that
I have
two
different
memories
and
that
I'm
encoding
something
in
time,
I'm
encoding
multiple
replicates
of something
in time,
to provide
a way of
checking to

make sure
that I'm
sending the
correct message.
so,
I can,
for example,
consider the
message to
be my
classical
memory,
and consider
the quantum
process as
a process
that tests
that memory.
So,
I remember
that I need
to do
experiment X
and I will
get result
Y.
I actually
do
experiment X
on this
quantum
channel and
see if I
get result
Y.
and if I
don't get
result Y,

then something's
wrong.
Either I
can't trust
the quantum
channel,
or I've
remembered
my preparation
conditions
incorrectly.
So,
I'm using
one memory
to check
the accuracy
of the
other.
And
this
problem
of using
something
about the
environment,
some channel
in the
environment,
to check
memory,
is absolutely
ubiquitous.
And we
use it all
the time.
So,
for example,
look around

your room
and your
room hopefully
looks more or
less the same
now that it
did at the
start of this
session.
And that's
very reassuring.
disturbing.
It reassures
you that
something's not
happening in
the environment
that's
dangerous and
disturbing.
And if you
couldn't do
that,
if your
environment was
constantly
changing,
and there
was no
familiarity
about the
environment,
you wouldn't
be able to
check your
memory.
You wouldn't
be able to

check your
sanity.
So,
even though we
don't think
about our
local environments
as effectively
an error
correcting code,
that's exactly
how we use
it.
We use
stegmergic
memory,
or memory
out there in
the environment,
as a way of
checking our
own memories,
and assuring
that our
memory is
okay.
So,
I talked a
little bit
about quantum
security.
Here's how
quantum security
system works.
Again,
it's a lock
protocol.
We have

some classical
signal that
can be
thought of as
a code,
or a
key,
or a
one-time
pad,
for example,
that encodes
a way of
doing
experiments.
And once
that's been
exchanged,
assuming it's
secure,
then we can
use the
quantum channel
to communicate
information.
And the
classical channel
has told us
what we have
to do
with that
quantum channel.
We have to
make certain
kinds of
measurements
at, say,
particular

clock times
as measured,
and we'll
get information
sent from
the past.
from the
sender.
And the
classical
channel protects
the quantum
channel from
adversarial agents
because anything
that an
adversarial agent
does to the
quantum channel,
if some
agent tries to
eavesdrop,
for example,
on the quantum
channel,
they'll disrupt
the quantum
channel,
and the
results of
our measurement
will not be
what's expected
given the
protocol that
we have
run.
So,

quantum
communication
is secured
by the fact
that interfering
with some
entangled state
actually destroys
the state,
so it's
detectable.
We can
always know
if someone
is trying
to disrupt
our communications,
which isn't
true if we
just have two
classical channels.
Charlie can
always intercept
a message
and then
recreate it,
and that's
not possible
with the
quantum
channel.
So,
let's
refold this
diagram
so that
we've taken
the external

time back
out of it,
and we
have this
picture where
the two
components of
Alice are
just interacting
via their
shared boundary
with their
environment,
Bob.
And now
we can see
that the
source of
redundancy
is not
time,
it's the
boundary
itself.
Just
sharing the
boundary
allows
Alice's
components,
A1 and
A2,
to put
copies of
the same
information
on different
places on

the boundary.
So,
the boundary
now becomes
a resource
for redundant
storage of
information.
So,
Alice 1 and
Alice 2 can
use different
parts of
the boundary
to check
what other
parts of the
boundary say,
to check the
information that's
encoded on other
parts of the
boundary.
So,
they can define
an error-correcting
code on the
boundary itself
using this
lock protocol.
So,
I just want
to leave you
with this
idea
that
using
classical and

quantum
channels together
allows us
to define
error-correcting
codes on
boundaries
that we share
as separate
agents.
And,
leave you
to think
about
what shared
measurements,
what shared
quantum reference
frames,
let us as
agents
see the
same things.
space.
And,
this is what
we'll talk
about next
time in
September.
We'll talk
specifically about
how space
provides us
with an
error-correcting
code.
And,

try to
understand
why space
is treated
as emergent
from something
else,
basically,
from
communication
in most
of quantum
gravity.
So,
thank you
very much.
I encourage
you to use
the interactive
QA and
to attend
the discussion
sessions.
And,
we'll see you
in September.
Thank you.
Awesome.
Chris,
thank you
for the
lecture.
Andrew,
do you have
any
remarks
or
questions?

questions?
Not right
now.
Thank you.
I'll just
ask a few
quick questions
and if anyone
has one
quickly in the
live chat.
So,
you mentioned
the tick
of the
clock.
And,
in active
inference,
we often
hold up
side-by-side
both the
discrete time
models and
the continuous
time models.
So,
do clocks
have to
tick
discreetly?
Or,
how do we
think about
continuous
time modeling?
In this

quantum setting,
they do have
to tick
discreetly
because one
has to be
able to
distinguish
Alice's
preparation
of the
cubits on
her boundary
from Bob's
measurements
of the
cubits on
the boundary.
And,
this process
can be
extremely fast
with respect
to any
kind of
sense of
macroscopic
time.
You know,
your eyes
are responding
to photons
in,
I think it's
hundreds of
femtoseconds.
So,
the

discrete
transitions
that are
effectively
time
measurements
for you
are at
that very
fast molecular
time scale,
much,
much faster
than
our
typical
clocks.
Now,
when we
try to
do very
fast timing,
we're
constructing a
faster clock
than that.
But,
it's a
clock that
we can
only observe
indirectly.
Awesome.
Okay.
Now,
this one
maybe connects
a little

bit to
application,
but with
all the
developments
happening
in theory,
some of
which you're
reviewing here,
and also
in practice
with quantum
computation,
and then
keeping in
mind this
distinction...
Sorry,
I've lost
you.
Back?
I've lost
you,
Daniel.
Okay.
We'll wait
until you're
back.
I can see
you.
No.
Okay.
Okay.
Just repeat
your question,
please.
Yeah,

so,
given these
rapid and
ongoing
developments
in theory
and practice
related to
quantum
information
science
and quantum
information
security,
how are
people
using
this
distinction
of the
classical
stigmatic
memory
and then
the quantum
cognitive
process
to anticipate
how to
develop
information
systems
that are
reliable
going
forward?
I don't
know of

any
instances
in which
thinking
about
quantum
cognition
has
entered
directly
into
any
design
of
communication
systems.
The
ability
of humans
to interact
with these
systems
is just
assumed.
So,
it's just
assumed
that the
user can
interact
with,
say,
a
computer
interface,
which is
the sort
of interface

that will
be provided
or is
provided
to any
quantum
computer.
So,
if you
think about
a quantum
computer,
a practical
quantum
computer,
it has
classical
interfaces
front and
back.
So,
in
thinking
about
interacting
with the
computer,
I'm giving
it some
classical
problem
statement,
for example,
and asking
for some
classical
answer,
and what

it's going
to give
me is a
probability
distribution
of classical
answers,
but I
can,
by running
the calculation
many times,
make that
probability
distribution
fairly tight.
so,
so,
yeah,
I don't
know of
anyone
talking
about
quantum
BCIs
at this
point.
It would
be an
interesting
question.
And,
was it
always
understood
these
sorts of

blind
spots and
axioms
associated
with
classical
communication?
was it
developed as
kind of like
a proposal
or like
a special
case and
more
unpacking
was left
for later?
Or are
these blind
spots things
that we're
now coming
into awareness
of given
what we're
learning in
the quantum
information
sciences?
I think
we're learning
how serious
they are,
but they
were certainly
recognized even
by Bohr

in his
1928
paper.
And,
that's a
very good
paper,
too.
It's called
The Present
State of
Quantum
Theory or
something like
that,
and it's
in Nature.
And,
it's available
on the
web
somewhere.
I have a
copy of
it.
But,
it was
written,
in a sense,
as Bohr's
summary of
the 1927
Solvay
conference,
which was
the conference
at which
everyone doing

quantum
theory at
that time,
got together
to talk
about their
results.
And,
in it,
Bohr
essentially
presents what
came to be
the Copenhagen
interpretation,
which I
think is
probably
unfairly
called an
interpretation.
It's sort
of the
Copenhagen
pragmatic
stance toward
quantum
theory.
And,
essentially,
what Bohr
says is
that if
we're going
to do
science,
we have
to assume

that people
can talk
to each
other.
And,
if we're
going to do
science,
we have to
assume that
there are
instruments
that we can
manipulate
and have
reasonable
confidence that
we know
what
manipulation
we've
done.
So,
if I turn
a knob
from one
to two,
I have to be
reasonably
confident that
I'm making a
particular change
that I can
understand.
and we have
to have
reasonable
confidence

that different
laboratories can
replicate the
same experiment.
And,
if we think
about any of
those assumptions
in purely
quantum theoretic
terms,
then all
sorts of
questions are
raised,
right?
Every quantum
system is
unique.
We have the
no-cloning
theorem.
So,
every scientific
instrument is
unique.
So,
there's no
such thing
as exactly
replicating an
experiment,
right?
That's a
coarse-grained
description of
what I've
done with a

completely unique
quantum system.
And my
brain, of
course, is
unique.
And yours is
too, and
everyone else's
is.
So,
talking about
sharing a
language is
essentially a
coarse-grained
assumption.
and what
the Copenhagen
stance is
all about
is coarse-graining
yielding
effective
classicality.
Or what that
really means is
coarse-graining
yielding a
description that
we can use
in the language
that we
understand.
So, these
are very,
very old
issues that

have been
recognized for
a long time.
But I
think in part
because of the
disaster of
shut up and
calculate,
they haven't
really been
talked about
in the
classroom.
Awesome.
I'll just
read short
questions from
the chat.
We can
pass them to
the discussion
section, but
there's a lot
of great
diverse
questions.
So, Alexi
wrote...
I think I've
lost you
again.
Oh!
The revenge
of Zoom.
Well, we
looked at the
chart earlier

where, like,
Bob was
talking to
Alice.
The
unidirectional
communication.
Okay, good?
Yeah.
Okay, okay,
okay.
The unidirectional.
The third
party is
recording it,
I'm sure.
Alexi wrote,
regarding the
environment for
being used for
error correction,
a symptom of
derealization in
psychological trauma
breakdown, we
don't recognize
the environment
around us and
it's scary?
Makes sense.
Makes sense.
All right, from
Bala.
Nice presentation.
I heard Chris
say, entangled
states are L-O-C-C
for a specific

reference frame.
Is this
entanglement
unitary or is
it related to
the numbers of
degrees of
freedom?
I think
the quick
answer would
be both.
If you have
a complex
system with
many degrees
of freedom,
some can be
entangled while
others aren't.
It depends
on what parts
of the state
are separable.
So, again,
entanglement
is just a
statement about
the
components
of a joint
state that
factor or
don't.
So, you can
have a joint
state that
looks separable

along some
dimensions but
doesn't
know the
dimensions.
And the other
big complication
here is that
whether a state
looks entangled
depends on the
reference frames
that you use
to measure it
with.
And if you
pick entangled
reference frames,
then what we
ordinarily think
of as
entangled states
become
separable.
So, you can
move entanglement
around.
Awesome.
Okay, from
Roland Rodriguez.
Is the
discrete
temporality
only within
the context
of a given
holographic
screen

interface?
With time
being bracketed
otherwise,
beyond interface
systems, is
continuity or
discretion kept
agnostic?
I think, if
I understand
the question
correctly, it's
yes.
If I don't
draw any
boundaries, I
just have the
joint state,
then the
evolution is
unitary.
And in a
sense, time
plays no
role.
And if
unitary
evolution is
perfectly
time-symmetric,
so time
actually drops
out altogether.
Awesome.
So, to talk
about time, we
have to draw

boundaries.
Awesome.
Time must have
a stop,
Huxley.
And I'll
ask one
last question
from the
appropriately
monikered
question
quanderer.
All
communication
is abductive
is a
sentiment that
I've been
using to
develop some
arguments.
Does that
scan for
you?
How do
we connect
what we're
discussing
here with
quantum to
abductive
logic?
Yeah, I
could go
along with
that.
I'd want a

lot more
detail.
All
communication
is essentially
analogical.
I would agree
with that,
yes.
Andrew?
going back
to the
picture,
the
scattering
picture,
so we
have so
far
everything
is topological
and it
certainly
induces,
you know,
breaking the
boundary in
different
sectors.
I suppose it
induces a
topology on
the boundary.
Now, I
have a couple
of questions.
As for the
geometry, and

this might
be getting
ahead,
is it an
emerging
property of
those channels
between
different
boundary
sectors?
And if
that's the
case, then
what does it
mean to have
a QRF for
space?
this is
really the
topic for
September,
but you
do point
out the
fact that
it's
something of
a chicken
and egg
situation.
The way
I think
of it is,
I suppose,
to choose
the chicken
side and

think about
what kind
of reference
frame does
the agent
have to
implement
to see
space on
their
boundary.
And the
reason I
think about
it that
way is
that's a
question I
want to be
able to
ask about
biological
systems.
And this
actually gets
into probably
what won't
be talked
about until
October.
But I
want to be
able to
ask,
essentially,
a phylogenetic
question of
what sorts

of organisms,
where in the
lineage from
LUCA, do
we find
systems that
are able to
assign spatial
coordinates to
their boundaries?
And what
coordinates come
first?
And I
suspect that
that sort
of discrete
location names
with no
metric come
first.
That a
system can
talk about,
for example,
this part of
their membrane
versus that
part of their
membrane,
without even
having any
ability to say
how far apart
they are.
that some
sort of
metric

representation
of this
angular space
comes later.
And that
a radial
space comes
even later
than that.
That's, at
this point,
speculation.
But I think
these are
questions that
we can end
up asking,
given the
right kind
of formalism
and the
right understanding
of what it
means to
be able to
make a
spatial
measurement.
So the
conjecture is
that the
metric would
emerge from
a discrete
network metric
of counting
ticks sort
of thing?

Yeah, I
think that's
correct.
And the
sort of
symmetries of
space that we
see are
very much
tied up with
object identity.
So if we
think about
something like
rotational
symmetry,
let's see,
oh,
I should
stop
screen
sharing.
So, you
know, I've
got this
object, right,
and I
can do
this, and
I say, oh,
you know, space
has this
rotational
symmetry.
But for me
to say that,
I have to
recognize that

this is the
same object,
right?
I have to
implement
object
persistence,
which we
all learn to
do when we're
nine months
old or
something.
but we
tend to
regard those
as separate
things as,
you know,
we learn
object persistence
because space
is rotationally
symmetric.
But we can
view it the
other way
around, that
space is
rotationally
symmetric for
me because I
implement these
constraints that
are called
object persistence.
Do you think
we can understand

phylogenetics as
a whole?
as a kind
of scattering
experiment,
like the
high-energy
LUCA event,
and then
dissipating and
propagating in
a very tangled
bank?
Well, I do
want to be able
to understand
phylogeny in
exactly the same
terms as
embryology.
And Mike
Levin and I
have published
papers about
that, that we
should be able
to view the
lineage of LUCA
in exactly the
same way that
we view the
lineage of the
zygote that
produced us.
So that's why
it's no coincidence
that during
pregnancy, the

development of
the embryo
mirrors evolution
in terms of the
parts that are
developed,
first?
Yeah, this is
the old idea
that phylogeny and
ontogeny are
related in some
way, which
from an embryonic
development point
of view,
they're certainly
analogous in
ways.
sorry, if I
may ask one
last question,
I think it
should be very
quick.
Regarding the
weak rotation,
I'm still a bit
unsure how it
ties to this
diagrammatic
notation that we
had with
quantum channels
and so on.
And there was
a note in the
second session,

we briefly talked
about the
weak rotation
being a
reversal in the
direction of
time.
Is there any
relation to
today's session?
Yeah, I,
whenever we
think about
the flow of
time as
driven by the
input-output
cycle, the
input-output
cycle for
agents separated
by a boundary
is always,
there's always a
sign change,
right?
my input is
your output
and vice
versa.
So, when
we think
about the
Bob's clock
ticking when
Alice does
something,
his direction

of time, in a
sense, points
with respect to
the boundary
opposite to
hers.
others.
But that's
only with
respect to
the boundary.
He sees
information flowing
in and she
sees information
flowing in.
That's the only
point being made
here about
the
wick rotation
as an
operation.
right.
Thank you.
Thank you,
Chris.
Very well
appreciated by
the viewers
and by us.
So, we'll
certainly look
forward to
reviewing the
Q&A and
everybody is
welcome and

encouraged to
join for the
participatory
discussion in
about two weeks
on Saturday.
So, till
next time.
All right.
Thank you.
Cheers.
to